

FUQUAN GAO

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Summary

Master of Information Technology student at The University of Melbourne, eager to secure a Full-Stack Software Engineer role to leverage background in distributed systems and cloud computing, Committed to building sovereign, high-impact AI solutions in Australia.

Detail-oriented professional with 7+ years of management experience and strong technical expertise in orchestrating Agile teams, optimising CI/CD pipelines, and deploying scalable solutions on AWS and Kubernetes

Education

Master of Information Technology

Mar, 2025 - Present

The University of Melbourne

Major: AI

Relevant Project Work

Full Stack Software Engineer

Rhombus AI, Sydney

Feb, 2026 - Present

Tech Stack: React, Next.js, TypeScript, Vite, Tailwind CSS, Python, Django, PostgreSQL, AWS, Jest, Terraform

- **Full-Stack Product Development:** Led the end-to-end delivery of multi-platform web applications. Leveraged React 18, Next.js 14, and Vite, alongside TypeScript and Tailwind CSS, to build accessible, responsive interfaces with high-fidelity animations using Radix UI and Framer Motion.
- **Auth & Data Layer:** Integrated Auth0, NextAuth, and Supabase for unified SSO, session management, and real-time data sync. Optimized performance using TanStack Query for server-state caching and Zustand/Redux for global client-state management.
- **Complex Interactions & Visualization:** Developed advanced workflow builders and canvas editors using React Flow (xyflow). Engineered data-dense dashboards with AG-Grid, Recharts, and Plotly, and implemented 3D visualizations via React Three Fiber to enhance data clarity.
- **Backend & Data Pipelines:** Developed custom Fivetran Connectors using Python (boto3, pandas) to process S3-hosted Excel data. Architected incremental sync and data cleaning pipelines to streamline the flow from raw data to the frontend.
- **Quality & Observability:** Established robust testing suites using Jest, Cypress, and Playwright with MSW for API mocking. Integrated Sentry for error tracking and PostHog for product analytics to drive data-informed stability and iterations.

- **Engineering Excellence:** Standardized code quality using ESLint, Prettier, Husky, and lint-staged. Utilized OpenAPI specifications and mock servers to enforce API contracts and streamline frontend-backend collaboration.

Full Stack Software Engineer

Oct, 2025 - Feb, 2026

MediaJira (SaaS Advertising Campaign Management Platform), Melbourne

Tech Stack: Next.js, TypeScript, Storybook, Django REST Framework, PyTest, OpenAPI 3.0.

- **Frontend Engineering:** Developed a Jira-style interactive Kanban board using Storybook components, implementing complex UI features including inline editing, dynamic column resizing, and sophisticated filtering, significantly enhancing task visualization capabilities.
- **State Machine Logic:** Architected a "Decision Engine" enforcing strict server-side state transitions (Draft -> Awaiting Approval -> Committed -> Reviewed). Implemented a logic-first frontend that strictly adheres to OpenAPI specifications and backend permission guards.
- **Backend Reliability:** Designed a comprehensive automated testing suite covering the entire decision lifecycle and RBAC permission boundaries. Ensured 0% side effects on invalid commit attempts by enforcing rigorous validation rules for signals and options.
- **API & Workflow:** Engineered the "Signals" management system with paired value/unit validation and delta tracking. Optimized data consistency by ensuring the UI strictly mirrors backend read-only states for committed records.

Full Stack Developer

Aug, 2025 - Nov, 2025

Health & Fitness Tracking App (Calories Group), Melbourne

Tech Stack: React Native, TypeScript, Kotlin, Spring Boot, Google GenAI, Firebase.

- **Mobile Engineering:** Designed a cross-platform app with reusable components and strict type-safe APIs. Integrated camera-based meal logging and geolocation services, delivering features like weight history visualization and restaurant recommendations.
- **Reactive Backend:** Built a **Kotlin + Spring Boot** backend using **Spring WebFlux** and coroutines. Exposed RESTful endpoints for diet/workout logs with centralized exception handling and request validation, ensuring high concurrency support.
- **AI Integration:** Integrated **Google GenAI** (Gemini) to provide chat-based nutrition coaching. Engineered analytics endpoints (e.g., diet/workout analysis, fat loss and muscle gain recommendations) consumed by a dedicated chat UI, enhancing user engagement.
- **Performance:** Optimized mobile performance via image compression and local storage (**AsyncStorage**). Encapsulated HTTP client logic to reduce code duplication and improve UX responsiveness.

Developer and Designer

Mar, 2025 - Jun, 2025

Cloud-Based Covid-19 Sentiment Analytics Platform, Melbourne

Tech Stack: Python, Kubernetes, Fission, Elasticsearch, Flask, Docker, NeCTAR Research Cloud.

- **Data Pipeline:** Designed a scalable pipeline collecting posts from Reddit and Mastodon; cleaned and normalized content before ingesting into **Elasticsearch** for downstream analysis.
- **Serverless Orchestration:** Built serverless crawler tasks using **Kubernetes** and **Fission**. Implemented Python handlers triggered via HTTP for both scheduled and ad-hoc data collection.
- **NLP Services:** Developed sentiment analysis services using **VADER** running as Fission functions. Automated the workflow to read raw posts, compute polarity scores, and bulk-write results to dedicated dashboard indices.
- **Visualization & API:** Delivered **Flask REST APIs** to query sentiment results and utilized **Jupyter notebooks** for cross-platform sentiment comparison (Reddit vs Mastodon), enabling exploratory analysis for researchers.

Employment

Technical Project Manager

May, 2024 - Feb, 2025

Louvresky Pty Ltd, Melbourne, VIC

- Led technical delivery and full SDLC for 3 cloud-native applications, ensuring architectural scalability while meeting strict on-time delivery and budget constraints
- Managed a cross-functional team of 6 developers and 2 QA engineers, resolving technical blockers and facilitating Agile ceremonies to streamline development workflows
- Architected automated testing workflows and enforced GitLab version control standards, slashing code deployment time by 40% and reducing production bugs by 15%
- Translated complex business requirements into technical specifications and user stories for the development team, achieving a 95% client satisfaction rate upon product release

Retail Manager

May, 2020 - Feb, 2024

Huawei Global Flagship Store, Shanghai, China

- Spearheaded daily operations for a flagship store, managing a team of 200+ staff members and overseeing store operations with monthly turnover exceeding \$5M
- Implemented data-driven CRM strategies, boosting customer engagement and increasing repeat visits by 20% year-over-year
- Led marketing campaigns and internal product training for 5 major product releases, ensuring 100% staff proficiency in technical specifications
- Streamlined warehouse logistics and auditing processes, minimising inventory discrepancies to less than 1% annually

Store Manager

Feb, 2017 - Dec, 2019

VideoEzy Ferntree Gully, Melbourne, VIC

- Supervised day-to-day store activities, including rostering and performance management for a team of 5 employees
- Revamped store layout and merchandising efficiency, directly contributing to a 25% increase in customer satisfaction scores
- Resolved complex customer escalations, maintaining a 4.8/5 star service rating for branch

Expert (Technical Specialist)

Apr, 2012 - Oct, 2015

Apple Retail Store, Doncaster, VIC

- Delivered exceptional technical support and product demonstrations to over 20 customers daily, routinely exceeding personal sales targets by 20%
- Mentored 10+ new employees on Apple product ecosystems and service standards
- Collaborated with cross-functional teams to streamline the "Genius Bar" appointment flow, reducing customer wait times by 10 minutes on average

Skills & Certifications

Programming Languages: Java, Python, C#, Swift, Objective-C, TypeScript, JavaScript (ES6+)

Frameworks & Libraries: Angular, Node.js, jQuery, Bootstrap, Webpack, Vite, Spring Boot

Cloud & DevOps: AWS, Docker, Kubernetes, Jenkins, GitLab CI/CD, GitHub Actions, Kafka

Databases: MySQL, Microsoft SQL Server, MongoDB, Elasticsearch

Certifications

AWS Certified DevOps Engineer - Professional (Issued Jul 2023)

Cisco Certified Network Associate (CCNA) (2024)

Microsoft Azure Fundamentals (AZ-900) (2024)

Google Cloud Digital Leader (2024)

Certified Kubernetes Administrator (CKA) - In Progress